COMP2006
Security Architectures and Cryptography

20 CREDIT MODULE

ASSESSMENT: 100% Coursework **W1: 30% Set exercise**
W2: 70% Report

MODULE LEADER: Dr Hai Van Dang

MODULE AIMS

- To develop an understanding of security architectures across a range of computing platforms (e.g. internet-enabled applications, embedded systems, cyber-physical systems).
- To critique the application of cryptography within existing implementations
- To analyse and discuss the role of privacy and its relationship with security.
- To design secure systems using knowledge of design principles, cryptography, and security controls.

ASSESSED LEARNING OUTCOMES (ALO):

1. Analyse and elicit security requirements given an IT scenario.
2. Apply knowledge of security architectures and design principles to design secure systems.
3. Develop appropriate cryptographic solutions to provide secure and privacy-enabled management of information.
4. Analyse the need for protection mechanisms within different scenarios, and explain their limitations in implementing protection.
5. Propose innovative solutions to security problems and critically evaluate their effectiveness

Overview

This document contains all the necessary information pertaining to the assessment of *COMP2006 Security Architectures and Cryptography* . The module is assessed via **100% coursework**, across two elements: *30% Set Exercises* and *70% Report*.

The sections that follow will detail the assessment tasks that are to be undertaken. The submission and expected feedback dates are presented in Table 1. All assessments are to be submitted electronically via the respective DLE module pages before the stated deadlines.

	Submission Deadline	Feedback
Set exercise (30%)	**3rd March 2026 (15:00)**	**31th March 2026**
Report (70%)	**5th May 2026 (15:00)**	**3rd June 2026**

Table 1: Assessment Deadlines

All assessments will be introduced in class to provide further clarity over what is expected and how you can access support and formative feedback prior to submission. **Whilst the assessment information is provided at the start of the module, it is not necessarily expected you will start this immediately** – as you will often not have sufficient understanding of the topic. The module leader will provide guidance in this respect.

Moderation

This assignment brief has been moderated in line with the university policy.

Moderator Name	Moderation Date
Rory Hopcraft	4 Feb 2026

Useful Assignment Information

Please see below some useful information regarding submissions for your modules.

Good Coding Practices

Like all things, no code is perfect. Just because your code compiles and runs does not mean it is perfect, there is always room for improvement. No code will achieve 100% of the available marks. Where code submission is required by a module, it will be assessed against the following criteria:

1) **Functionality**
    a. Does your code meet all the specified requirements of the assignment?
    b. Does your code behave correctly across typical and edge-case scenarios?
    c. Does you code have appropriate error handling that does not leading to it crashing or undefined behaviour?
2) **Efficiency**
    a. Is your code optimised in terms of performance and resource use?
    b. Does you code handle functions, variables, data calls efficiently?
    c. Is there any unnecessary repetition, complexity or processing reducing efficiency within your code?
3) **Readability**
    a. Is your code logically structured and easy to read?
    b. Have you maintained standard formatting conventions like indentations, spacing and naming, within your code?
    c. Are variables, functions, and classes clearly named and purposeful?
4) **Documentation**
    a. Are there helpful comments explaining non-obvious parts of your code?
    b. Do your comments document your process, development or rationale clearly?
    c. Could someone unfamiliar with the code understand the approach you have taken?

Use of Generative AI for Creating Code

Each assignment element for each module will have a clear indication of the permitted level of use of AI (solo, assisted or partnered) including the generation of code. Please ensure that you **read and understand the permitted use**. If you are unsure of a particular use, please reach out to the Module Leader and ask.

We strongly encourage you to read and explore beyond the core content delivered within the modules. However, this must be done correctly following academic process. Wherever code is drawn from an outside source (e.g. you have not written it yourself), regardless of whether this is AI generated or from an online repository (GitHub, Stack Overflow etc) **you must reference the original source**. This can be done in your documentation as comments or part of your write ups. Students found to be utilising code without referencing could face an academic offence. **If in doubt speak to your Module Leader.**

Versioning

A critical part of a development (software or academic) is versioning. Keeping a number of iterations over the course of development ensures you always have backups to fall back on. It also provides clear demonstration of the development process you implemented.

Using online/cloud-based storage solutions and repositories provide additional peace of mind, alongside being industry practice. The university provide OneDrive to all students which offers inbuilt version history for Microsoft products. GitHub is the university recommended repository for versioning code, which includes great integration with a number of IDEs.

Submitting Code

It is vital that you confirm that all code that you submit is in the **correct format** and **compiles correctly** or **runs without error**. If you clean up your code before submission, please ensure that all dependencies (libraries, functions and variables) are included so that the marker can compile your code.

We can only mark what has been submitted. If the code does not run correctly, it is not our responsibility to spend time error handling to award you marks.

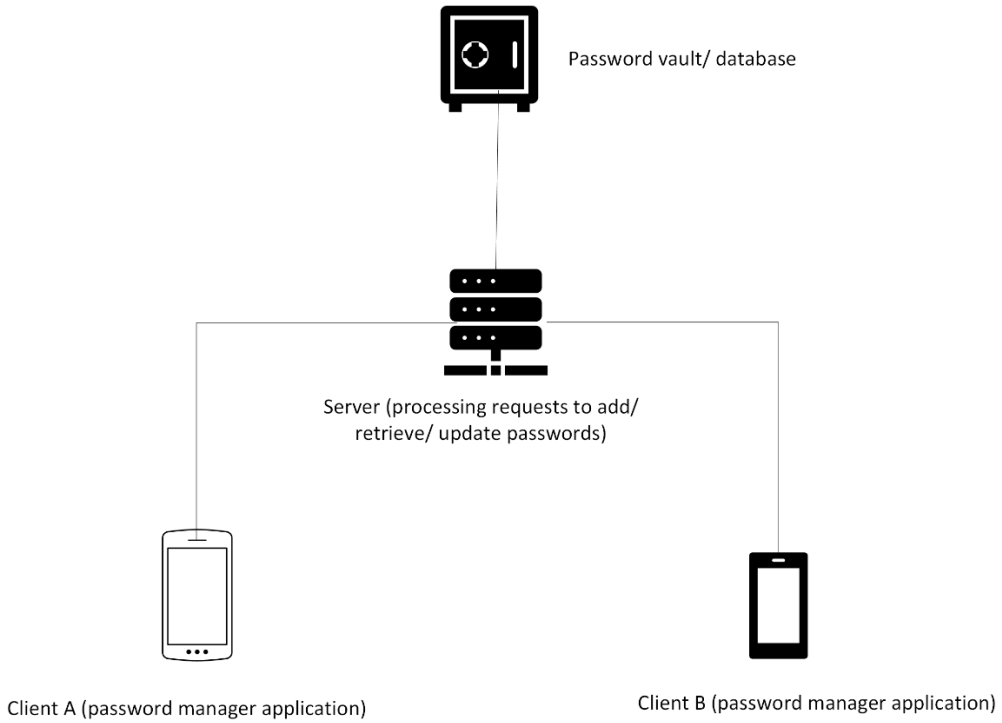
It is also important to **show your working**. Tidying up your code is a critical part of coding practices, but if you remove the workings that provides you with the required output, we cannot see how you got there.

In a worst-case scenario, if you remove a key part of your working, and on compiling your code it gives an error or different result, we have no way of confirming what went wrong, or indeed if you fabricated those outputs.

Please note, just because a piece of code has given the wrong output, it does not mean it should be deleted. Seeing these attempts with comments documenting your attempt allows us to understand where you may have made an error and provide feedback that addresses this. In some cases, you could also be awarded marks for the attempt.

Assessment 1: Set exercise

Task:
Password manager applications (such as NordPass, 1Password) provide users with support to manage their multiple passwords or passkeys. Company X is developing a new password manager application for mobile phone. The system architecture is as follows:



Client is the password manager application on a user's device which enable the users to request to add a new password, view/ update/ delete a password (e.g. password of a website, password of an application). Server is a robust backend that processes the clients' such requests. The client' passwords are stored securely at the password vault/ database, accessible via the client's master password.

As an independent security architect, your task is to compete for the tender by analysing the security threats and proposing a security architecture for the system. Within the proposal, you are requested to:

1. Identify and explain what security attacks against the security objectives defined in CIA triad and AAA framework can happen in the system **(40 marks)**.
2. Propose a cryptographic protocol to protect the **confidentiality** of **the passwords stored at the password vault/ database** (this cryptographic protocol will be used as a security mechanism) and **describe** how this protocol works when a client requests to add a new password. **Explain** what attacks identified in question (1) and any additional attacks the proposed protocol can prevent and how **(60 marks).**

Submission Requirements:

- 1500-word limit
- Report submitted at as single PDF document to DLE
- Appendix with completed Generative AI Declaration

Acceptable level of generative AI tool use in this assessment element

The acceptable level of GenAI use for this element is detailed with the allowed uses listed below. If you have any questions, please contact the module leader.

Assisted Work	You are permitted to use generative AI tools in an assistive role.	☒

Table 2: Acceptable Level of generative AI use

Inline with the indicated acceptable level of AI use above, the following uses are acceptable:

- **A1 – Idea Generation and Problem Exploration**
- **A2 - Planning & Structuring Projects**
- **A4 – Research Assistance**

Any use of AI in your work must be declared within your documentation. You **must also include a signed Generative AI Declaration as an appendix to your submission**. The declaration form can be found on DLE (or Here on the Programme Page). This form will not be included in any word count associated with this assignment.

Assessment Criteria:

The report MUST be structured as specified in the following table. Please be very careful and pay close attention to these instructions. The following marking scheme will be used for this part of the assessment:

Content	Details	Marks
Section 1 – Question 1	Identify and explain what security attacks against the security objectives defined in CIA triad and AAA framework can happen in the system. The explanation should provide enough details to understand the meaning of the attacks in the given system instead of generic meaning, how it can happen, what components are affected and what can be sequences.	**40**
Section 2 – Question 2	The description of the proposed protocol should include the assumptions (e.g. what information each component needs to know or store), message flow, the chosen algorithm(s), and actions involved in the protocol (e.g. what each component in the system (client A, client B, server) would need to do in each step). You do NOT need to implement the protocol. The description on how this protocol works when a client requests to add a new password should include the actions of each relevant components (e.g. what each component needs to do). The analysis should explain what attacks identified in section (1) and any additional attacks it can prevent and how	**60**
Appendix - Generative AI Declaration	Student Declaration of AI Tool use in this Assessment (download from DLE) Please do not include your name or signature in the form because the marking will be anonymous.	

An illustration of the feedback template is presented in Table 3.

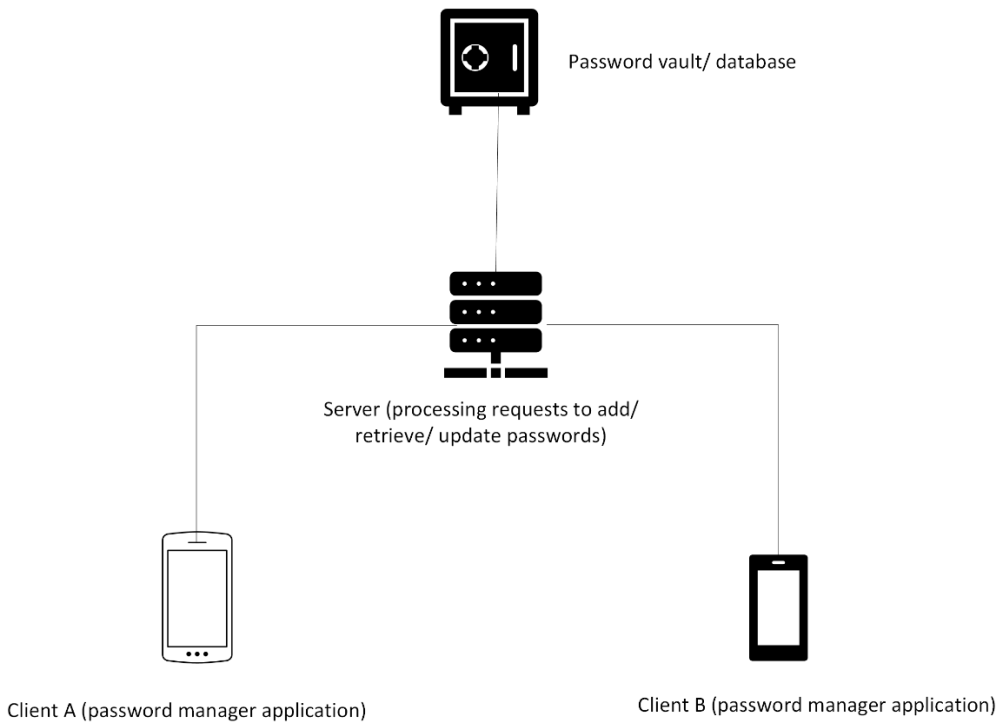
Threshold Criteria (these are indicative only):

Criteria	Fail (<40%)	3rd (40%+)	2:2 (50%+)	2:1 (60%+)	1st (70%+)	Grade
Section 1	No or irrelevant security attacks have been identified	A few relevant security attacks with explanation have been provided	A good list of relevant security attacks with clear explanation have been provided	A very good list of relevant security attacks with good explanation have been provided	An excellent list of security attacks with excellent explanation have been provided	**/40**
Section 2	No or irrelevant protocol has been proposed No explanation and analysis have been provided	A relevant protocol has been proposed with little explanation and analysis	A good protocol has been proposed with proper explanation and analysis	A very good protocol has been proposed with good explanation and analysis	An excellent protocol has been proposed with excellent explanation and analysis	**/60**
Feedback/Overall	*Additional feedback*					**/100**

Table 3: Feedback Template for Assessment 1

Assessment 2: Report

Task:
Password manager applications (such as NordPass, 1Password) provide users with support to manage their multiple passwords or passkeys. Company X is developing a new password manager application for mobile phone. The system architecture is as follows:



Client is the password manager application on a user's device which enable the users to request to add a new password, view/ update/ delete a password (e.g. a website password, an application password). Server is a robust backend that processes the clients' such requests. The client's passwords are stored securely at the password vault/ database, accessible via the client's master password.

As an independent security architect, you are now invited to compete for the tender in the next phase. The company requests you to

1. Propose, describe and analyse **two** security mechanisms for authentication: (1) one security mechanism for authentication between the client and server, and (2) one security mechanism for authentication between the server (which processing requests to add/retrieve/update passwords) and the password vault/ database (which is another server). Your proposal should apply and **follow the security design principles** as much as possible (60 marks).
2. Implement the security mechanism for **authentication between** the server (which processing requests to add/retrieve/update passwords) and the password vault/ database (which is another server) described in question (1) with explanation. Show the evidence of implementation plus an input/ output example (40 marks).

Submission Requirements:

- 3500-word limit
- Report submitted at as single PDF document to DLE
- Appendix with completed Generative AI Declaration

Acceptable level of generative AI tool use in this assessment element

The acceptable level of GenAI use for this element is detailed with the allowed uses listed below. If you have any questions, please contact the module leader.

Assisted Work	You are permitted to use generative AI tools in an assistive role.	☒

Table 2: Acceptable Level of generative AI use

Inline with the indicated acceptable level of AI use above, the following uses are acceptable:

- **A1 – Idea Generation and Problem Exploration**
- **A2 - Planning & Structuring Projects**
- **A4 – Research Assistance**

Any use of AI in your work must be declared within your documentation. You **must also include a signed Generative AI Declaration as an appendix to your submission**. The declaration form can be found on DLE (or Here on the Programme Page). This form will not be included in any word count associated with this assignment.

Assessment Criteria:

The report MUST be structured as specified in the following table. Please be very careful and pay close attention to these instructions. The following marking scheme will be used for this part of the assessment:

Content	Details	Marks
Section 1 – Question 1	Propose, describe and analyse **TWO** security mechanisms as required. Your proposal should apply and follow the security design principles as much as possible. The description of the mechanism should include the chosen approach/ algorithm/ technique, the information flow, actions (e.g. what each component in the system (client, server, vault/ database) would need to do), etc. The analysis should explain what security design principles have been applied in your proposal, how and the benefits. Instruction: 1. The user would need to log into the client app to use the service. Therefore, you could also consider user authentication for authentication between the client and server. Meanwhile, the server does not hold the user log-in information, therefore, user authentication would not be suitable for authentication between server-password vault/ database. 2. The security mechanisms could either your own proposed protocol based on the learnt cryptographic concepts or a standard protocol used in the industry.	**60**

	In case you propose a solution based on a standard protocol used in the industry, you should explain in detail the relevant steps of the protocol which provide authentication. For both cases, the description of the mechanism should describe detailed steps, which algorithms are used in each step (when appropriate), which techniques are used (when appropriate), the information flow (which information each party/component needs to send or receive or store), actions (e.g. what action/ computation each relevant component/party would need to do), explain which party/component is authenticating and which one is authenticated, why the authentication works and how it can prevent the impersonation attack, etc.	
Section 2 - Question 2	Provide the screenshots with explanation to demonstrate the implementation of the **ONE** security mechanism for **authentication between** the server (which processing requests to add/retrieve/update passwords) and the password vault/ database (which is another server) described in section 1. The screenshots should show the details of implementation, the input and output example. Instruction: 1. If your proposed solution is your own and based on the learnt cryptographic concepts, your implementation could demonstrate the steps that you describe and explain in question 1. 2. If your solution is based on a standard protocol in industry, your implementation could demonstrate the detailed relevant steps that you describe and explain in question 1.	**40**
Appendix - Generative AI Declaration	Student Declaration of AI Tool use in this Assessment (download from DLE) Please do not include your name or signature in the form because the marking will be anonymous.	

An illustration of the feedback template is presented in Table 3.

Criteria	Fail (<40%)	3rd (40%+)	2:2 (50%+)	2:1 (60%+)	1st (70%+)	Grade
Section 1-question 1	No or irrelevant security mechanisms have been provided. No or irrelevant security design principles have been applied. No or irrelevant explanation and analysis have been provided.	At least one relevant security mechanism has been provided. A few security design principles have been applied. Some explanation and analysis have been provided.	The required relevant security mechanisms have been provided with clear description A good list of security design principles have been applied. Good explanation and analysis have been provided.	The required relevant security mechanisms have been provided with good description A very good list of security design principles has been applied. Very good explanation and analysis have been provided.	The required relevant security mechanisms have been provided with excellent description A excellent list of security design principles has been applied. Excellent explanation and analysis have been provided.	**/60**
Section 2-question 2	No or irrelevant implementation has been provided. No or irrelevant explanation and evidence has been provided	Some relevant implementation has been provided with evidence and some explanation.	A good implementation has been provided with evidence and good explanation	A very good implementation has been provided with evidence and very good explanation	An excellent implementation has been provided with evidence and excellent explanation	**/40**
Feedback/Overall	*Additional feedback*					**/100**

Table 4: Feedback Template for Assessment 2

Acceptable levels of AI use:

The table below provides the acceptable use categories for GenAI. Each assessment element may allow different uses. Please check the brief for each element carefully to see what uses are allowed.

Solo Work	**S1 - Generative AI tools have not been used for this assessment.**
Assisted Work	**A1 – Idea Generation and Problem Exploration** Used to generate project ideas, explore different approaches to solving a problem, or suggest features for software or systems. Students must critically assess AI-generated suggestions and ensure their own intellectual contributions are central.
	A2 - Planning & Structuring Projects AI may help outline the structure of reports, documentation and projects. The final structure and implementation must be the student's own work.
	A3 – Code Architecture AI tools maybe used to help outline code architecture (e.g. suggesting class hierarchies or module breakdowns). The final code structure must be the student's own work.
	A4 – Research Assistance Used to locate and summarise relevant articles, academic papers, technical documentation, or online resources (e.g. Stack Overflow, GitHub discussions. The interpretation and integration of research into the assignment remain the student's responsibility.
	A5 - Language Refinement Used to check grammar, refine language, improve sentence structure in documentation not code. AI should be used only to provide suggestions for improvement. Students must ensure that the documentation accurately reflects the code and is technically correct.
	A6 – Code Review AI tools can be used to check comments within the code and to suggest improvements to code readability, structure or syntax. AI should be used only to provide suggestions for improvement. Students must ensure that the code accurately reflects their knowledge and is technically correct.
	A7 - Code Generation for Learning Purposes Used to generate example code snippets to understand syntax, explore alternative implementations, or learn new programming paradigms. Students must not submit AI-generated code as their own and must be able to explain how it works.
	A8 - Technical Guidance & Debugging Support AI tools can be used to explain algorithms, programming concepts, or debugging strategies. Students may also help interpret error messages or suggest possible fixes. However, students must write, test, and debug their own code independently and understand all solutions submitted.
	A9 - Testing and Validation Support AI may assist in generating test cases, validating outputs, or suggesting edge cases for software testing. Students are responsible for designing comprehensive test plans and interpreting test results.
	A10 - Data Analysis and Visualization Guidance AI tools can help suggest ways to analyse datasets or visualize results (e.g. recommending chart types or statistical methods). Students must perform the analysis themselves and understand the implications of the results.
	A11 - Other uses not listed above Please specify:
Partnered Work	**P1 - Generative AI tool usage has been used integrally for this assessment** Students can adopt approaches that are compliant with instructions in the assessment brief. Please Specify:

General Guidance

Extenuating Circumstances

There may be a time during this module where you experience a serious situation which has a significant impact on your ability to complete the assessments. The definition of these can be found in the University Policy on Extenuating Circumstances here:
https://www.plymouth.ac.uk/student-life/your-studies/essential-information/exams/exam-rules-and-regulations/extenuating-circumstances

Plagiarism

All of your work must be of your own words. You must use references for your sources, however you acquire them. Where you wish to use quotations, these must be a very minor part of your overall work.

To copy another person's work is viewed as plagiarism and is not allowed. Any issues of plagiarism and any form of academic dishonesty are treated very seriously. All your work must be your own and other sources must be identified as being theirs, not yours. The copying of another persons' work could result in a penalty being invoked.

Further information on plagiarism policy can be found here:

Plagiarism: https://www.plymouth.ac.uk/student-life/your-studies/essential-information/regulations/plagiarism

Examination Offences: https://www.plymouth.ac.uk/student-life/your-studies/essential-information/exams/exam-rules-and-regulations/examination-offences

Turnitin (http://www.turnitinuk.com/) is an Internet-based 'originality checking tool' which allows documents to be compared with content on the Internet, in journals and in an archive of previously submitted works. It can help to detect unintentional or deliberate plagiarism.

It is a formative tool that makes it easy for students to review their citations and referencing as an aid to learning good academic practice. Turnitin produces an 'originality report' to help guide you. To learn more about Turnitin go to:
https://help.turnitin.com/new-links.htm?Highlight=guide

The university also had Draft Coach available on the online version of Microsoft Word. Draft Coach is a Turnitin plugin and provides instant feedback on how to address citation issues, grammar mistakes and matches with Turnitin's database.

Referencing

The University of Plymouth Library has produced an online support referencing guide which is available here: http://plymouth.libguides.com/referencing.

Another recommended referencing resource is Cite Them Right Online; this is an online resource which provides you with specific guidance about how to reference lots of different types of materials.